

C Programming Book

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Chapter 1

Introduction to C

- Comparison between English language and C programming language

English Language	C Language
1. Alphabets (a, b, c, ...)	1. Keywords (32)
2. Words (More than 10K)	2. Operators (45)
3. Sentence (Grammer makes this)	3. Separators (14)
	4. Constants

- Constants

Item	value	C datatype
Integer	5 , -5 , 0	int
real	5.12 , 0.39 , 5.00 , -3.25	float
charater	'a' , '5'	char

- Reason for the name float but why not real?
 - 100. 231452156
 - * . is decimal point
 - * 231452156 : This keep on changing So float
- Primary / Primitive / Fundamental / Basic data types in C
 - Data is a meaningfull information
 - Datatype is a different kind of data like int , float and Char

Format Specifier	Data Type	Description
%c	char	Prints a single character
%s	char *	Prints a string
%d	int	Signed decimal integer
%i	int	Signed integer (same as %d in printf)
%u	unsigned int	Unsigned decimal integer
%hd	short int	Signed short integer
%hu	unsigned short int	Unsigned short integer
%ld	long int	Signed long integer
%lu	unsigned long int	Unsigned long integer
%lld	long long int	Signed long long integer
%llu	unsigned long long int	Unsigned long long integer
%f	float	Floating-point number (printf promotes float to double)
%lf	double	Double-precision floating-point (scanf uses %lf for double)
%Lf	long double	Long double floating-point

Format Specifier	Data Type	Description
%e / %E	float, double	Scientific notation
%g / %G	float, double	Shortest representation of %f or %e
%x	unsigned int	Hexadecimal integer (lowercase)
%X	unsigned int	Hexadecimal integer (uppercase)
%o	unsigned int	Octal integer
%p	void *	Pointer address
%%	—	Prints % character
%zu	size_t	Unsigned size type
%td	ptrdiff_t	Pointer difference type
%jd	intmax_t	Largest signed integer type
%ju	uintmax_t	Largest unsigned integer type

1.1 List of all Operators

Operator	Description	Associativity
() [] . -> ++ --	Parentheses or function call Brackets or array subscript Dot or Member selection operator Arrow operator Postfix increment/decrement	left to right
++ -- + - ! ~ (type) * & sizeof	Prefix increment/decrement Unary plus and minus not operator and bitwise complement type cast Indirection or dereference operator Address of operator Determine size in bytes	right to left
* / %	Multiplication, division and modulus	left to right
+ -	Addition and subtraction	left to right
<< >>	Bitwise left shift and right shift	left to right
< <= > >=	relational less than/less than equal to relational greater than/greater than or equal to	left to right
== !=	Relational equal to or not equal to	left to right
&&	Bitwise AND	left to right
^	Bitwise exclusive OR	left to right
 	Bitwise inclusive OR	left to right
&&	Logical AND	left to right
 	Logical OR	left to right
? :	Ternary operator	right to left
= += -= *= /= %= &= ^= = <<= >>=	Assignment operator Addition/subtraction assignment Multiplication/division assignment Modulus and bitwise assignment Bitwise exclusive/inclusive OR assignment	right to left
,	comma operator	left to right

Figure 1.1: Operators

1.2 Assignment Operator

- It requires 2 arguments
- Left side arg must be variable
- Right side arg can be a variable / constant / expression
- Left side value is called l-value

```
1 // Not valid
2 a = 5
3 a = ;
4   = 10;
5 10 = 20;
6
7 // Valid
8 a = 5;
9 b = a;
10 c = 2 + 3;
```

- Every expression is replaced with constant

```
1 a = 5;
2 b = -3;
3 c = 9;
4
5 a = b; // a = -3
6 b = -c; // b = -9
7 a = -b; // a = 9
8
9 -c = a; // Error because l-value is expression not a variable
```

- In C language each and every statement ends with `;`
- Semicolon is called statement terminator (or) end of statement
- Why `.` is not used as terminator?
 - because `.` is used as decimal point in float number
- What is l-value error?
 - On the left hand side of assignment operation we should provide a variable but by mistake if we provide constant (or) expression then we get l-value required error.

1.3 Arithmetic Operators

```
1 // Example 1: Expression evaluation
2 a = 2 + 3;
3 a = 2 + 3 + 4;
4 a = 6 - 2 + 3 - 4 + 1;
5 a = 2 + 3 * 5;
6 a = (2 + 3) * 5;
7 a = 2 * 3 + 3 * 2;
```

Operation on	Result
int with int	int
int with float	float
float with int	float
float with float	float

```
1 // Example 2:
2 a = 5 / 2; // 2
3 a = -5 / 2; // -2
4 a = 5 / -2 // -2
```

```

5  a = -5 / -2 // -2
6
7  a = 5.0 / 2; // 2.5
8  a = 5 / 2.0; // 2.5
9  a = -5 / 2.0 // -2.5
10 a = 2 / 5 // 0
11 a = 2 / -5 // 0

```

```

1  // Example 3:
2
3  a = 17 / 3 / 2;
4  /*
5  a = 5 / 2
6  a = 2
7  */
8
9  a = 15 / 2 * 3;
10 /*
11 a = 7 * 3
12 a = 21
13 */
14
15 a = 9 * / 4;
16 /*
17 a = 27 / 4
18 a = 6
19 */
20
21 a = 6 * 3 / 4 * 5;
22 /*
23 a = 18 / 4 * 5
24 a = 4 * 5
25 a = 20
26 */

```

Division	Result of division	Modulus	Result of modulus
+ / +	+	+ % +	+
+ / -	-	+ % -	+
- / +	-	- % +	-
- / -	+	- % -	-

- Floating point numbers cannot used in modulus only integer is acceptable

```

1  // Example 1
2
3  a = 5 / -2; // -2
4  a = 5 % -2; // 1

```

```

1  // Example 2
2
3  a = 3.5 * 2 % 7; // Compilation error

```

- The result of modulus operation is remainder
- Result sign is same as numerator sign
 - If numerator is +ve then result is +ve
- Result sign does not depend on denominator
- Condition 1 : $x \% y == 0$ means then we can say that x is multiple of y
- Condition 2 : Numerator % 10 gives last digit of numerator

- Condition 3 : When numerator is smaller then denominator then result is numerator
- Condition 4 : When numerator is equal to denominator then result is zero

1.4 C-Tokens

- Smallest individual unit in C language is called C-Token
- C-Token are
 - Keywords
 - Operators
 - Separators
 - Constants
 - Identifiers
- There can be 'N' no. of space / Tabs / Newlines between two C-Tokens
- Program should be in good indentation

```
1 void main()
2 {
3     printf("Hello");
4 }
```

- The above program has following C-Tokens
 - Keyword : void
 - Operator : () ()
 - Separator : { ; }
 - Constant : "Hello"
 - Identifiers : main printf
- Total 11 C-Tokens are there in above program

1.5 printf function

- It will print 1st arg on screen
- 1st arg must be in pair of double quotes
- if more than one arg, then every arg must be separated by commas

```
1 printf("Hello");
2 // Hello
3
4 printf("    Hello    ");
5 // ...Hello...
6 // (Spaces are retained)
7
8 printf("Hello%dadc%d", 10, 20);
9 // Hello10abc20
10
11 printf("%d %d %d", 10, 20, 30);
12 // 10 20 30
13
14 printf("%d%d%d", 10, 20, 30);
15 // 102030
16
17 printf("%d,%d,%d", 10, 20, 30);
18 // 10,20,30
19
20 printf("%d %d %d", 10, 20);
21 // 10 20 <GV/Junk>
22
23 printf("%d %d %d", 10);
```

```

24 // 10 GV GV
25
26 printf("%d %d %d");
27 // GV GV GV
28
29 printf("%d %d %d", 10, 20, 30, 40);
30 // 10 20 30
31
32 printf("%d %f %c", 10, 3.75, 'a');
33 // 10 3.75 a
34
35 printf("%d", 5.5);
36 // GV
37
38 printf("%f", 5);
39 // GV
40
41 printf("%d", 5 + 2);
42 // 7
43
44 printf("5 + 2");
45 // 5 + 2
46
47 printf("%d + %d", 5 + 2);
48 // 7 + GV
49
50 printf("%d * %d = %d", 5, 2, 5+2);
51 // 5 * 2 = 7
52
53 printf("%d * %d = %d", 5, 2, 5*2);
54 // 5 * 2 = 10
55
56 printf("%f", 5/2);
57 // GV
58
59 printf("%d", 5/2);
60 // 2
61
62 printf("%f", 5.0 % 2);
63 // Error
64
65 printf("%d", -5 % -2);
66 // -1
67
68 printf("Hello", "Hai", "Bye");
69 // Hello
70
71 printf("Hello"Hai"Bye");
72 // HelloHaiBye
73
74 printf("Hello");
75 // Error
76
77 printf("Hello");
78 // Hello
79
80 printf("Hello");
81 //Error

```

```

82
83 printf("%d %d %d", "%d %d", 10, 20, 30, 40, 50);
84 // GV 10 20
85
86 printf("10, 20", "%d");
87 // 10, 20
88
89 int ts = 7500;
90 printf("ts");           // ts
91 printf(ts);             // Error
92 printf("Total sal = %d", ts); // Total sal = 7500
93 printf("%d + 1000", ts);  // 7500 + 1000
94 printf("%d", ts + 1000); // 8500

```

1.6 Variable definition Rules

- Variable definition ways,

```

1 // Method 1:
2 int a;
3 int b;
4 int c;
5
6 // Method 2:
7 int a, b, c;

```

- In the above two style 1st style is recommended because,
 - Subsequent change in variable type in future is easy
 - Writing and updating a comment is easy
- **Variable definition:** Compiler allocates memory for that variable

```

1 int a;
2 int b = 10;

```

- **Variable declaration:** Compiler doesn't allocate memory for that variable

```

1 extern int a;

```

Examples:

```

1 void main()
2 {
3     int a; // Always say datatype to a variable
4     a = 5;
5     printf("%d", a);
6 }
7
8 // Output
9 // 5
10
11 void main()
12 {
13     int i;
14     float f;
15     i = 5;
16     f = 5.5;
17     printf("%d %f", i, f);
18 }
19
20 // Output

```

21 // 5 5.5

- In C programming we can assign any type of data to any type of variable. Internally in computer the compiler will take care.
- In printf function only suitable data is given to format specifier else garbage value is printed.

```
1 void main()
2 {
3     int i;
4     i = 5.5;
5     printf("%d", i); // 5
6 }
7
8 void main()
9 {
10    float f;
11    f = 5;
12    printf("%f", f); // 5.0
13 }
14
15 void main()
16 {
17    int i;
18    i = 5 / 2;
19    printf("%d", i); // 2
20 }
21
22 void main()
23 {
24    float f;
25    f = 5 / 2;
26    printf("%f", f); // 2.0
27 }
28
29 void main()
30 {
31    int i;
32    i = 5.0 / 2;
33    printf("%d", i); // 2
34 }
35
36 void main()
37 {
38    float f;
39    f = 5.0 / 2;
40    printf("%f", f); // 2.5
41 }
```

- More examples

```
1 /***** Valid definition *****/
2
3 int a;
4 a = 5; // Assignment
5
6 int a = 5; // Initialization
7
8 int a = 18 / 3 / 5;
9
```

```

10 int a = 2 + 5;
11
12 int a = 7.65;
13
14 int a, b, c;
15 a = b = c = 10; // Valid
16
17 int a, b, c = a = b = 10;
18
19 int a = a;
20
21 int a1, b1, c1;
22
23 int avgofsalary; // Valid but not readable
24
25 int avg_of_salary;
26
27 int a, A;
28
29 char way2ms;
30
31 char INT;
32
33 int abcdefghijklmnopqrst;
34
35 int _;
36
37 char _1, _2;
38
39 /***** Not a valid *****/
40 int a = b = c = 10;
41
42 int 1a, 1b, 1c;
43
44 int avg of salary; // Not a valid. No white space allowed
45
46 int if;
47
48 int a + b;
49
50 int a, a; // Not a valid. same name not allowed
51
52 float 1606y2;
53
54 int 42shared;
55
56 char `a`, `b`, `c`;

```

- Rules for naming a valid variable Names:
 - An Identifier can contain
 - * a - z
 - * A - Z
 - * 0 - 9
 - * _ (underscore)
 - An Identifier name start with alphabet (or) underscore
 - No whitespace, operator are allowed
 - No keyword, but act as a part of variable
 - * Example: `int intfloat`

- No restriction in length of variable name, but if length is very big then readability is less. Hence programmers recommendation is max of 15 characters

```
1 int pwm_analog_read;
2 int pwm_analog_write;
3 int pwm_digital_read;
4 int pwm_digital_write;
```

1.7 Relational Operators

Operator	Name	Description
<code>==</code>	Equal to	Checks whether two operands are equal
<code>!=</code>	Not equal to	Checks whether two operands are not equal
<code>></code>	Greater than	Checks whether left operand is greater than right operand
<code><</code>	Less than	Checks whether left operand is less than right operand
<code>>=</code>	Greater than or equal to	Checks whether left operand is greater than or equal to right operand
<code><=</code>	Less than or equal to	Checks whether left operand is less than or equal to right operand

```
1 a = 5 + 2; // a is int
2
3 b = 1.5 * 3; // b is float
4
5 c = 4 > 3; // c and d is boolean (but not available in C)
6 d = 9 < 7;
```

- The use of this operator
 - To establish relation between two numbers and perform comparison
 - Result of relation operation is either `true` or `false`
 - `true` and `false` are boolean datatype but C does not support that hence
 - * `true` = Non-zero
 - * `false` = zero

```
1 void main()
2 {
3     int a;
4     a = 4 > 3 > 2;
5     printf("%d", a); // 0
6 }
```

- In Maths,
 - $a > b$ and $b > c$ makes $a > c$. So $a > b > c$
 - $4 > 3$ and $3 > 2$. So $4 > 3 > 2$
- In English, Lets take below two sentences
 - I am watching movie
 - I am eating popcorn
 - Now, Lets combine two statements blindly
 - * I am watching movie eating popcorn (meaning less)
 - So, I am watching movie `and` eating popcorn (meaningful now)
- In C Language
 - $4 > 3 > 2$ makes result in `0` even if we think in mathematically and grammatically
 - So, $(4 > 3) \&\& (3 > 2)$ makes result `1`

```
1 a = 4 + 3 > 2 + 5;
2   = 7 > 2 + 5
3   = 7 > 7
```

```

4   = 0
5
6   a = 6 > 3 + 2 < 8
7   = 6 > 5 < 8
8   = 1 < 8
9   = 1
10
11  a = 6 * 3 >= 5 * 2 + 8 >= 17 + 8 <= 6 * 4 == 8 * 5 != 5 * 3 * 2 + 10 != 40;
12  // Based on the Associativity of operator all are executed from left to right
13  // and last operator is !=. So result always will be 1 or 0

```

1.8 Logical Operator

- If we are combining 2 or more relation operator directly then result is unexpected
- To overcome this C people have introduced logical operators
- Thus use of logical operation is to combine two (or) more relational statement and to get compound statement

Operator	Name	Description
&&	Logical AND	Returns true if both conditions are true
	Logical OR	Returns true if at least one condition is true
!	Logical NOT	Reverses the logical state of operand

A	B	A && B	A B	!A	!B
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0



- Whenever we are having 'N' condition and if we are depend on all 'N' condition at that time we are going for AND logic
- When there are 'N' condition and if we are depend on any 1 condition that time we gone for OR logic
- NOT is used in negative test condition

Chapter 2

Flow control

2.1 if Conditional Construct

```
1 // (1)
2
3 if( condition ) {
4     // (2) if body;
5 }
6
7 // (3)
```

- If condition is `True` then (1), (2) and (3) will execute.
- If condition is `False` then (1) and (2) will execute.

```
1
2 void main()
3 {
4     printf("A");
5     printf("B");
6
7     if(3 > 2) {
8         printf("C");
9         printf("D");
10    }
11
12    printf("E");
13    printf("F");
14 }
15
16 // Output
17 // ABCDEF
```

```
1
2 void main()
3 {
4     printf("A");
5     printf("B");
6
7     if(3 > 20) {
8         printf("C");
9         printf("D");
10    }
11
12    printf("E");
13    printf("F");
14 }
15
16 // Output
17 // ABEF
```

- If there is no `if` body / `else` body then compiler will assume the first semicolon as a part of `if` body

```

1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 2)
6         printf("C");
7         printf("D");
8         printf("E");
9         printf("F");
10 }
11
12 // Output
13 // ABCDEF

```

```

1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 20)
6         printf("C");
7         printf("D");
8         printf("E");
9         printf("F");
10 }
11
12 // Output
13 // ABDEF

```

- In C programming `;` alone is valid.
- The purpose of the `;` is to provide a delay to next instruction.
- The line that contains only `;` is called Null statement, Empty statement, Dummy statement.

```

1 // Valid program
2 void main()
3 {
4     ;
5 }

```

```

1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 2);
6         printf("C");
7         printf("D");
8         printf("E");
9         printf("F");
10 }
11
12 // Output
13 // ABCDEF
14

```

```

1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 20);
6         printf("C");
7         printf("D");
8         printf("E");
9         printf("F");
10 }
11
12 // Output
13 // ABCDEF
14

```

```

1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 2);           // Compiler will execute this with meaningless
6     {                   // Dummy body start
7         printf("C");
8         printf("D");
9     }                   // Dummy body end
10    printf("E");
11    printf("F");
12 }
13

```

```
14 // Output
15 // ABCDEF
```

- In C programming, Non-zero values are consider as True and zero is consider as False

```
1
2 void main()
3 {
4     printf("A");
5     printf("B");
6     if(20)
7     {
8         printf("C");
9         printf("D");
10    }
11    printf("E");
12    printf("F");
13 }
14
15 // Output
16 // ABCDEF
```

```
1
2 void main()
3 {
4     printf("A");
5     printf("B");
6     if(0)
7     {
8         printf("C");
9         printf("D");
10    }
11    printf("E");
12    printf("F");
13 }
14
15 // Output
16 // ABEF
```

```
1
2 void main()
3 {
4     printf("A");
5     printf("B");
6     if( )
7     {
8         printf("C");
9         printf("D");
10    }
11    printf("E");
12    printf("F");
13 }
14
15 // Output
16 // Compilation Error
```

- Interview based program

```
1 void main()
2 {
3     int a = 10;
4     int b = 20;
5     int max;
6
7     /* To find max of two integer using if */
8     if(a > b) {
9         max = a;
10    }
11
12    if(b > a) {
13        max = b
14    }
15
16    /* WAP to find max of 2 integer using only one if */
17    max = b;
18    if(a > b) {
19        max = a;
20    }
21
22    /* WAP to find max of 2 integer using 1 if and without 3rd variable */
23    if(a > b) {
24        b = a;
25    }
26    printf("MAX = %d", b);
27
28    /* WAP to find max of 2 integer without using any if without 3rd variable */
29    max = (a > b) * a + (b < a) * b;
30
31    /* WAP to find max of 2 integer without using any condition */
32    max = (a + b + abs(a - b)) / 2;
```

33 }

2.2 if-else Conditional Construct

```
1 // (1)
2
3 if( condition ) {
4     // (2) if body
5 } else {
6     // (3) else body
7 }
8
9 // (4)
```

- If condition is `True` then (1), (2) and (4) will execute.
- If condition is `False` then (1), (3) and (4) will execute.
- Whenever condition is true do some specific job. When condition is false do some other specific task. At that time we are going for `if-else` conditional construct.
- `else` keyword must be immediately placed after `if` body.
- If some statement is placed between `if` body and `else` then that is error.

```
1 void main()
2 {
3     printf("A");
4     printf("B");
5     if(3 > 2)
6     printf("C");
7     printf("D");
8     else
9     printf("E");
10    printf("F");
11    printf("G");
12    printf("H");
13 }
14
15 // Output
16 // Compilation Error
```

Program to find max of 2 integers

```
1 if(a > b) {
2     max = a;
3 } else {
4     max = b;
5 }
```

Program to find max of 3 integers

```

1  if((a > b) && (a > c)){
2      max = a;
3  } else {
4      if(b > c) {
5          max = b;
6      } else {
7          max = c;
8      }
9  }

```

```

1  if(a > b) {
2      if(a > c) {
3          max = a;
4      } else {
5          max = c;
6      }
7  } else {
8      if(b > c) {
9          max = b;
10     } else {
11         max = c;
12     }
13 }

```

Examples

```

1
2  void main() {
3      int a = 10;
4      if(a == 5) {
5          printf("Hello%d", a);
6      } else {
7          printf("Hai%d", a);
8      }
9  }
10
11 // Output
12 // Hai5

```

```

1
2  void main() {
3      int a = 10;
4      if(a == 10) {
5          printf("Hello%d", a);
6      } else {
7          printf("Hai%d", a);
8      }
9  }
10
11 // Output
12 // Hello10

```

- Assignment operation has 2 actions
 - Assign value to variable
 - Replace the expression with assigned value
- Why this behaviour?

```

1  int a1, b1, c1;
2  c1 = b1 = a1 = 10; // This results in all variables assigned to value 10
3
4  int a2;
5  float b2;
6
7  a2 = b2 = 3.5 // a2 = 3, b2 = 3.5
8  b2 = a2 = 3.5 // a2 = 3, b2 = 3.0

```

```

1 void main() {
2     int a = 10;
3     if(a = 0) {
4         printf("Hello%d", a);
5     } else {
6         printf("Hai%d", a);
7     }
8 }
9
10 // Output
11 // Hai0
12

```

```

1 void main() {
2     int a = 10;
3     if(a = 1) {
4         printf("Hello%d", a);
5     } else {
6         printf("Hai%d", a);
7     }
8 }
9
10 // Output
11 // Hello1
12

```

```

1 void main() {
2     int a = 10;
3     if(a = 0.75) {
4         printf("Hello%d", a);
5     } else {
6         printf("Hai%d", a);
7     }
8 }
9
10 // Output
11 // Hai0
12

```

More analysis

```

1 // What if the below conditions are in if condition?
2
3 a == 10; // Hello10
4 a == 5; // Hai10
5 a == 1; // Hai10
6 a == 0; // Hai10
7 a == 10.5; // Hai10
8 a == 10.0; // Hello10
9 a == 0.75; // Hai10
10
11 a = 10; // Hello10
12 a = 5; // Hello5
13 a = 1; // Hello1
14 a = 0; // Hai10
15 a = 10.5; // Hello10
16 a = 10.0; // Hello10
17 a = 0.75; // Hai10

```

```

1 a = 10;
2 b = 20;
3 c = a && b = 30;
4
5 printf("%d %d %d", a, b, c);
6
7 // Output
8 // Compilation Error: l-value required

```

```

1 a = 10;
2 b = 20;
3 c = a && (b = 30);
4
5 printf("%d %d %d", a, b, c);
6
7 // Output
8 // 10 30 1

```

- 2.3 conditional Operator
- 2.4 while loop
- 2.5 Nested while Loops
- 2.6 For loop
- 2.7 Do while Loop
- 2.8 Break
- 2.9 Continue
- 2.10 goto
- 2.11 Switch
- 2.12 Summary of Flow Control
- 2.13 Increment & Decrement Operators_1
- 2.14 Increment & Decrement Operators_2
- 2.15 Number System
- 2.16 Intro To Data Types
- 2.17 Signed char Data Type
- 2.18 Unsigned char Data Type
- 2.19 Short int Data Type
- 2.20 Long int Data Type
- 2.21 Floatingpoint Datatypes
- 2.22 Sizeof Operator

Chapter 3

Bitwise Operators

Operator	Name	Description
&	Bitwise AND	Sets bit to 1 if both bits are 1
	Bitwise OR	Sets bit to 1 if at least one bit is 1
^	Bitwise XOR	Sets bit to 1 if bits are different
~	Bitwise NOT / One's Complement	Inverts all bits
«	Left Shift	Shifts bits to left
»	Right Shift	Shifts bits to right

3.1 One's Complement

!	~
Output is 1 or 0	output is integer
!3.14 is 0	~3.14 is error

```
1 void main()
2 {
3     int a;
4     a = ~0;
5     printf("%d", a);
6 }
7
8 /*
9 if int datatype = 2 bytes
10
11 0 = 0000 0000 0000 0000 (result = 0)
12 ~0 = 1111 1111 1111 1111 (result = -1)
13 */
```

- If MSB = 0, then number is +ve and it is original data
- If MSB = 1, then number is -ve and number is in 2's complement.

3.1.1 Binary(2's complement) to decimal

- If MSB = 0, Convert normally from binary to decimal.
- If MSB = 1,
 1. Take 2's complement again
 2. Convert to decimal
 3. Add negative sign

Example

```
1 /*
2 Binary(2's complement) = 1111 1111 1111 0101
3 original data           = 0000 0000 0000 1010
4                               +1
5 -----
6 0000 0000 0000 1011 = 11 (consider this as negative)
7 */
```

```
1 /*
2 ~5
3 ~(0000 0000 0000 0101)
4 1111 1111 1111 1010
5           | |
6           V V
7         4+1 = 5
8           |
9           V
10        5 + 1 = 6 (consider as negative)
11
12 ~5 = -6
13 */
```

```
1 /*
2 ~9
3 ~(0000 0000 0000 1001)
4 1111 1111 1111 0110
5           | |
6           V V
7         8 +1 = 9
8           |
9           V
10        9 + 1 = 10 (consider as negative)
11
12 ~9 = -10
13
14 So,
15 ~0 = -1
16 ~5 = -6
17 ~9 = -10
18 ~37 = -38
19 ~~6 = 5
20 ~~3107 = 3106
21 ~~10906 = 10905
22 ~3.14 = Error (Complement on float is error)
23 */
```

3.1.2 Summary



1. Add 1
2. Change the sign of result

3.2 Left shift Operator

```
1  /*
2  5 << 1 = 0000 0000 0000 0101
3           0000 0000 0000 1010 (shift by 1) = 10
4
5  5 << 2 = 0000 0000 0000 0101
6           0000 0000 0001 0100 (shift by 2) = 20
7
8  -5 << 1 = 1111 1111 1111 1011
9           1111 1111 1111 0110 (shift by 1) = -10
10
11 -6 << 2 = 1111 1111 1111 1010
12           1111 1111 1110 1000 (shift by 2) = -24
13 */
```

3.2.1 Summary



$a \ll b$ is equal to $a * 2^b$

3.3 Right shift Operator

```
1  /*
2  5 >> 1 = 0000 0000 0000 0101
3           0000 0000 0000 0010 = 2
4           |
5           V
6           This position is filled by the value of MSB because signed
7
8  11 >> 2 = 0000 0000 0000 1011
9           0000 0000 0000 0010 = 2
10          ||
11          VV
12          This position is filled by the value of MSB because signed
13
14 -9 >> 1 = 1111 1111 1111 0111
15          1111 1111 1111 1011 = -5
16          |
17          V
18          This position is filled by the value of MSB because signed
19
20 -6 >> 2 = 1111 1111 1111 1010
21          1111 1111 1111 1110 = -2
22          ||
23          VV
24          This position is filled by the value of MSB because signed
25 */
```

3.3.1 Summary

1.

$$5 >> 1 = \frac{5}{2^1} = 2$$

2.

$$11 \gg 2 = \frac{11}{2^2} = 2$$

3.

$$-9 \gg 1 = \sim \left(\frac{\sim -9}{2^1} \right) = \sim \left(\frac{8}{2} \right) = \sim (4) = -5$$

4.

$$-6 \gg 2 = \sim \left(\frac{\sim -6}{2^2} \right) = \sim \left(\frac{5}{4} \right) = \sim (1) = -2$$



$$a \gg b = \begin{cases} \frac{a}{2^b}, & \text{if } a \text{ is positive} \\ \left(\frac{\sim -a}{2^b} \right) & \text{if } a \text{ is negative} \end{cases}$$

- For signed data gap is filled with MSB
- For unsigned data gap is filled with 0
- Shifting like `10 << 40` , `81 >> 100` , `10 << -2` , `8 >> -3` are machine dependant

3.4 Bitwise AND, OR, XOR

AND	XOR	OR
<code>x & 1 => x</code>	<code>x ^ 1 => !X (Toggle)</code>	<code>X 1 => 1 (Set)</code>
<code>x & 0 => 0 (Clear)</code>	<code>X ^ 0 => X</code>	<code>X 0 => X</code>

- `X` is don't care

3.4.1 Summary



1. Set a bit, use OR
2. Clear a bit, use AND
3. Toggle a bit, use XOR

3.5 Problem solving

3.5.1 Check if the nth bit is set

```

1 bool is_bit_set(uint32_t x, uint8_t n)
2 {
3     return (x >> n) & 1U;
4     /* OR: return (x & (1U << n)) != 0; */
5 }

```

3.5.2 Set the nth bit

```

1 uint32_t set_bit(uint32_t x, uint8_t n)
2 {
3     return x | (1U << n);
4 }

```

3.5.3 Clear the nth bit

```
1 uint32_t clear_bit(uint32_t x, uint8_t n)
2 {
3     return x & ~(1U << n);
4 }
```

3.5.4 Toggle the nth bit

```
1 uint32_t toggle_bit(uint32_t x, uint8_t n)
2 {
3     return x ^ (1U << n);
4 }
```

3.5.5 Count number of set bits

```
1 uint8_t count_set_bits(uint32_t x)
2 {
3     uint8_t count = 0;
4     while (x) { /* loop runs exactly as many times as there are set bits */
5         x &= x - 1; /* x-1 flips all bits from LSB up to (and including) the
6                     lowest set bit, so AND-ing zeroes that bit out.
7                     e.g. x    = 1010 1000
8                        x-1  = 1010 0111
9                        x&x-1 = 1010 0000 <- one set bit gone */
10        count++;
11    }
12    return count; /* result = number of 1-bits in original x */
13 }
```

3.5.6 Check if a number is a power of 2

```
1 bool is_power_of_2(uint32_t n)
2 {
3     /* n=0 excluded: 0 is not a power of 2 */
4     return (n != 0) && ((n & (n - 1)) == 0);
5 }
```

3.5.7 Swap two numbers using XOR

```
1 void xor_swap(uint32_t *a, uint32_t *b)
2 {
3     if (a != b) { /* guard against aliasing */
4         *a ^= *b;
5         *b ^= *a; /* *b = original *a */
6         *a ^= *b; /* *a = original *b */
7     }
8 }
```

3.5.8 Find first set bit in a number

```
1 /* Shift right until LSB is 1, counting each step.
2    * 0(position of lowest set bit) - worst case 0(32). */
3 int first_set_bit_manual(uint32_t x)
4 {
5     if (!x) return -1;
```

```

6
7     int pos = 0;
8     while ((x & 1U) == 0) {      /* while LSB is 0... */
9         x >>= 1;                /* ...shift right (discard LSB) */
10        pos++;                  /* ...and count the step */
11    }
12    return pos;                  /* pos where (x >> pos) & 1 is first true */
13 }
14
15 /* isolation trick
16  *
17  * -x in two's complement = ~x + 1
18  * ~x flips all bits, +1 then ripples carry rightward until it sets
19  * exactly the position of the original lowest set bit.
20  * Result is a VALUE with only that one bit set, not the position.
21  *
22  * e.g.  x      = 0000 1010 0 (LSB at bit 1)
23  *      -x      = 1111 0110 0
24  *      x & -x = 0000 0010 0 <-> isolated LSB as a value      */
25 uint32_t isolate_lsb(uint32_t x)
26 {
27     return x & (-x);            /* returns the bit VALUE, not position */
28 }

```



$x = 0000\ 1010\ 1000$ (lowest set bit at position 3) $x - 1 = 0000\ 1010\ 0111$ (borrow flips bits 0..3) $x \& -x = 0000\ 0000\ 1000$ (isolates ONLY bit 3 as a value)

3.5.9 Find last set bit in a number

```

1  /* Keep shifting x right until it becomes 0.
2  * Each shift discards one bit from the top.
3  * The number of shifts needed = position of the highest set bit.
4  *
5  * e.g.  x = 0110 0100 (decimal 100)
6  *      shift 1 -> 0011 0010   pos = 1
7  *      shift 2 -> 0001 1001   pos = 2
8  *      ...
9  *      shift 6 -> 0000 0001   pos = 6
10 *      shift 7 -> 0000 0000   loop exits, return pos = 6      */
11 int last_set_bit_shift(uint32_t x)
12 {
13     if (!x) return -1;
14
15     int pos = 0;
16     while(x) {
17         x = x >> 1      /* shift right; loop exits when x becomes 0 */
18         pos++;          /* count each shift, final value = MSB position */
19     }
20     return pos;
21 }
22
23 /* Binary search / divide-and-conquer
24  *
25  * Narrows down the MSB position in exactly 5 steps (log2 of 32).

```

```

26  * Checks whether any bit exists in the upper half; if yes, shift
27  * the entire value down by that half and accumulate the offset.
28  *  $O(\log 32) = O(5)$  – no loop dependency on the actual bit position.
29  *
30  * Step sizes: 16 -> 8 -> 4 -> 2 -> 1
31  int last_set_bit_bsearch(uint32_t x)
32  {
33      if (!x) return -1;
34
35      int pos = 0;
36
37      if (x & 0xFFFF0000U) { pos += 16; x >>= 16; } /* bit in upper 16? */
38      if (x & 0x0000FF00U) { pos += 8; x >>= 8; } /* bit in upper 8? */
39      if (x & 0x000000F0U) { pos += 4; x >>= 4; } /* bit in upper 4? */
40      if (x & 0x0000000CU) { pos += 2; x >>= 2; } /* bit in upper 2? */
41      if (x & 0x00000002U) { pos += 1; x >>= 1; } /* bit in upper 1? */
42
43      return pos;
44  }

```

3.5.10 Reverse bits of 32-bit number

```

1  /*
2  * Reverse all 32 bits of an unsigned integer.
3  * e.g.  0b 1011 0100 0000 0000 0000 0000 0000 0001
4  *       0b 1000 0000 0000 0000 0000 0000 0010 1101 (result)
5  */
6
7  /* Divide-and-conquer with bitmasks (portable,  $O(\log 32)$ )
8  *
9  * Core idea: repeatedly swap the two halves of the number,
10 * each time at half the previous granularity.
11 *
12 * 5 stages: 16-bit halves -> bytes -> nibbles -> bit-pairs -> bits
13 *
14 * Mask anatomy (memorise these – they come up in other problems too):
15 *
16 * 0xFFFF0000 = 1111 1111 1111 1111 | 0000 0000 0000 0000 (upper half)
17 * 0x0000FFFF = 0000 0000 0000 0000 | 1111 1111 1111 1111 (lower half)
18 *
19 * 0xFF00FF00 = 1111 1111 | 0000 0000 | 1111 1111 | 0000 0000 (odd bytes)
20 * 0x00FF00FF = 0000 0000 | 1111 1111 | 0000 0000 | 1111 1111 (even bytes)
21 *
22 * 0xF0F0F0F0 = 1111 | 0000 | 1111 | 0000 | ... (odd nibbles)
23 * 0x0F0F0F0F = 0000 | 1111 | 0000 | 1111 | ... (even nibbles)
24 *
25 * 0xCCCCCCCC = 11 | 00 | 11 | 00 | ... (odd bit-pairs)
26 * 0x33333333 = 00 | 11 | 00 | 11 | ... (even bit-pairs)
27 *
28 * 0xAAAAAAAA = 1010 1010 ... (odd single bits)
29 * 0x55555555 = 0101 0101 ... (even single bits)
30 *
31 * Pattern: each stage's two masks are bitwise complements of each other.
32 * Each stage shifts by exactly half its working block size.
33 */
34 uint32_t reverse32(uint32_t x)
35 {

```

```

36  /* Stage 1: swap upper 16 bits with lower 16 bits
37  *   before: [A B C D | E F G H | I J K L | M N O P]
38  *   after:  [I J K L | M N O P | A B C D | E F G H] */
39  x = ((x & 0xFFFF0000U) >> 16) |
40      ((x & 0x0000FFFFU) << 16);
41
42  /* Stage 2: swap odd bytes with even bytes within each 16-bit half
43  *   before: [I J K L | M N O P | A B C D | E F G H]
44  *   after:  [M N O P | I J K L | E F G H | A B C D] */
45  x = ((x & 0xFF00FF00U) >> 8) |
46      ((x & 0x00FF00FFU) << 8);
47
48  /* Stage 3: swap odd nibbles with even nibbles within each byte
49  *   each byte: [HI LO] -> [LO HI] */
50  x = ((x & 0xF0F0F0F0U) >> 4) |
51      ((x & 0x0F0F0F0FU) << 4);
52
53  /* Stage 4: swap odd bit-pairs with even bit-pairs within each nibble
54  *   each nibble: [b3 b2 | b1 b0] -> [b1 b0 | b3 b2] */
55  x = ((x & 0xCCCCCCCCU) >> 2) |
56      ((x & 0x33333333U) << 2);
57
58  /* Stage 5: swap every odd bit with its even neighbour
59  *   final swap of adjacent bits - completes the full reversal */
60  x = ((x & 0xAAAAAAAAU) >> 1) |
61      ((x & 0x55555555U) << 1);
62
63  return x;
64 }
65
66 /* Naive loop
67 *
68 * Extract LSB of x each iteration, push it into result from the MSB end.
69 * 0(32) - always 32 iterations regardless of bit pattern. */
70 uint32_t reverse32_loop(uint32_t x)
71 {
72     uint32_t result = 0;
73     int bits = 32;
74
75     while (bits--) {
76         result = result << 1; /* shift result up*/
77         result = result | (x & 1U); /* OR in LSB of x */
78         x = x >> 1; /* discard the LSB just consumed */
79     }
80
81     return result;
82 }

```

3.5.11 Swap even and odd bits

```

1  uint32_t swap_even_odd_bits(uint32_t x)
2  {
3      uint32_t odd_bits = x & 0xAAAAAAAAU; /* isolate bits at pos 1,3,5,7,...
4                                              zero out all even-position bits */
5
6      uint32_t even_bits = x & 0x55555555U; /* isolate bits at pos 0,2,4,6,...
7                                              zero out all odd-position bits */

```

```

8
9     return (odd_bits >> 1)          /* slide odd-position bits DOWN by 1
10                                     pos 1->0, 3->2, 5->4, 7->6, ... */
11     | (even_bits << 1);           /* slide even-position bits UP by 1
12                                     pos 0->1, 2->3, 4->5, 6->7, ... */
13 }

```

3.5.12 Check if the number has only one bit set

If the number is **power of 2** then that number has only one bit set.

3.5.13 Clear all the bits from MSB to nth bit pos

```

1  /*
2  * Clear all bits from the MSB down to position n (inclusive).
3  * Keep bits [n-1 .. 0] untouched.
4  *
5  * Visual:
6  * bit pos: 31 30 29 ... n+1  n | n-1  n-2 ... 1  0
7  * before:   ?  ?  ? ... ?  ? | ?  ?  ... ?  ?
8  * after:    0  0  0 ... 0  0 | ?  ?  ... ?  ?  <- upper cleared
9  *
10                                     ^^^^^^^^^^^^^^^^^^^^^^^^^
11                                     these bits preserved exactly
12 *
13 * e.g. x = 0xFF (1111 1111), n = 4
14 *      clear bits [7..4], keep bits [3..0]
15 *      result = 0x0F (0000 1111)
16 */
17 /* Method 1: Lower-bits mask (preferred – one expression)
18 *
19 * Key insight: (1U << n) - 1 produces exactly n ones in the low positions.
20 *
21 * n = 4:
22 * 1U << 4      = 0000 0000 0001 0000 (1 at position n)
23 * (1U << 4) - 1 = 0000 0000 0000 1111 (all ones below position n)
24 *
25 * AND-ing with this mask:
26 * - bits [n-1..0] : masked with 1 -> preserved unchanged
27 * - bits [31..n]  : masked with 0 -> forced to 0 (cleared) */
28
29 uint32_t clear_msb_to_n(uint32_t x, uint8_t n)
30 {
31     return x & ((1U << n) - 1U); /* keep only the lower n bits */
32     /*
33      * (1U << n) - 1U builds the "keep" mask at compile time if n is
34      * a constant – zero runtime cost on any architecture.
35      */
36 }
37
38 /* Method 2: Complement + shift (shows understanding of bit width)
39 *
40 * Build a mask of ones for the upper bits, then invert and AND.
41 *
42 * ~0U          = 1111 1111 1111 1111 1111 1111 1111 1111
43 * ~0U << n      = 1111 1111 1111 1111 1111 1111 1111 0000 (n=4)
44 * ~(~0U << n)  = 0000 0000 0000 0000 0000 0000 0000 1111 <- same mask

```

```

45  *
46  * Identical result to Method 1, but makes the complement relationship
47  * explicit – useful when explaining to an interviewer.          */
48
49  uint32_t clear_msb_to_n_v2(uint32_t x, uint8_t n)
50  {
51      return x & ~(~0U << n);          /* ~0U = all ones; shift left clears low n;
52                                         invert again -> low n ones = keep mask */
53  }

```

3.5.14 Extract a bit field from a register

3.5.14.1 Given start bit position and len

```

1  uint32_t extract_bitfield(uint32_t reg, uint8_t start, uint8_t width)
2  {
3      return (reg >> start)           /* Step 1: shift field to bit 0      */
4          & ((1U << width) - 1U);     /* Step 2: mask off everything above */
5  }

```

3.5.14.2 Given start bit position and end bit position

```

1  uint32_t extract_bits(uint32_t value,
2                        uint8_t start,
3                        uint8_t end)
4  {
5      uint32_t width = end - start + 1;
6      uint32_t mask = (1U << width) - 1;
7      return (value >> start) & mask;
8  }

```

3.5.14.3 Given mask and position

```

1  uint32_t extract_with_cmsis_style(uint32_t reg,
2                                    uint32_t mask,
3                                    uint8_t pos)
4  {
5      return (reg & mask) >> pos;
6  }

```

3.5.15 Insert a bit field into a register

3.5.15.1 Given start bit position and width

```

1  uint32_t insert_bitfield(uint32_t reg,
2                          uint32_t field_val,
3                          uint8_t start,
4                          uint8_t width)
5  {
6      uint32_t mask = ((1U << width) - 1U) << start;
7
8      reg &= ~mask;          /* Clear field region      */
9      reg |= (field_val << start) & mask; /* Insert new field value */
10
11     return reg;
12 }

```


3.5.15.2 Given start bit position and end bit position

```
1 uint32_t insert_bits(uint32_t reg,  
2                     uint32_t field_val,  
3                     uint8_t start,  
4                     uint8_t end)  
5 {  
6     uint32_t width = end - start + 1;  
7     uint32_t mask = ((1U << width) - 1U) << start;  
8  
9     reg &= ~mask;           /* Clear target bits */  
10    reg |= (field_val << start) & mask; /* Insert field */  
11  
12    return reg;  
13 }
```

3.5.15.3 Given mask and position

```
1 uint32_t insert_with_cmsis_style(uint32_t reg,  
2                                 uint32_t field_val,  
3                                 uint32_t mask,  
4                                 uint8_t pos)  
5 {  
6     reg &= ~mask;           /* Clear field */  
7     reg |= (field_val << pos) & mask; /* Insert shifted value */  
8  
9     return reg;  
10 }
```


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